Today

- One-dimensional systems
- The infinite square well
- Finite square well
- Delta function potential

2 One-dimensional systems

Now, we begin constructing the wavefunctons for a system containing a single particle of mass m in one dimension. Different systems correspond to different potentials. We will be considering only potentials that are time-independent. The wavefunctions and the corresponding energies are then obtained by solving the time-independent Schrödinger equation in one dimension,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x). \tag{2.0.1}$$

Once we find the wavefunction $\psi_E(x)$ of energy E, its time dependence follows easily,

$$\psi_E(x,t) = \psi_E(x)e^{-\frac{i}{\hbar}Et}. \tag{2.0.2}$$

2.1 Boundary conditions

The boundary conditions we impose depend on the potential. Suppose we are given a generic potential $V(x)$. Consider a point $x = a$ on the $x-$axis. We want to investigate whether the wavefunction $\psi(x)$ and its first derivative $\psi'(x)$ are continuous or not across $x = a$. First, rewriting the Schrödnger equation as

$$\psi''(x) = \frac{2m}{\hbar^2}(V(x) - E)\psi(x), \tag{2.1.1}$$

we note that the continuity or discontinuity of $\psi''(x)$ follows the continuity or discontinuity of the potential $V(x)$.

For the first derivative, and considering a particle of finite energy E,

$$\lim_{\epsilon\to 0}\left(\psi'(a+\epsilon) - \psi'(a-\epsilon)\right) = \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\psi''(x)dx = \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\frac{2m}{\hbar^2}(V(x) - E)\psi(x)dx$$

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$$= \lim_{\epsilon \to 0} \frac{2m}{\hbar^2} \int_{a-\epsilon}^{a+\epsilon} V(x)\psi(x)dx, \tag{2.1.2}$$

Now if $V(x)$ has a finite jump at $x = a$, this integral vanishes and $\psi'(x)$ is continuous across $x = a$. On the other hand, if $V(x)$ has an infinite jump, similar to the delta function, the integral will be nonzero but finite, and $\psi'(x)$ is discontinuous across $x = a$.

For the wavefunction itself,

$$\lim_{\epsilon \to 0} (\psi(a+\epsilon) - \psi(a-\epsilon)) = \lim_{\epsilon \to 0} \int_{a-\epsilon}^{a+\epsilon} \psi'(x)dx. \tag{2.1.3}$$

Now whether $V(x)$ has a finite or infinite jump at $x = a$, this integral vanishes, since $\psi'(x)$ could have only a finite jump across $x = a$. Therefore, $\psi(x)$ is always continuous across $x = a$.

We summarize the boundary conditions as follows

1. When the potential $V(x)$ has a finite jump at $x = a$, both $\psi(x)$ and $\psi'(x)$ are continuous across $x = a$.
2. When the potential $V(x)$ has an infinite jump at $x = a$, $\psi(x)$ is continuous but $\psi'(x)$ is discontinuous across $x = a$.

Furthermore, the wavefunction must vanish at $x = \pm\infty$ for a normalizable wavefunction.

2.2 Infinite square well

Consider a particle of mass m in the potential

$$V(x) = \begin{cases} 0, & 0 \le x \le a \\ \infty, & x < 0 \text{ and } x > a. \end{cases}$$

The potential is shown in the figure below.

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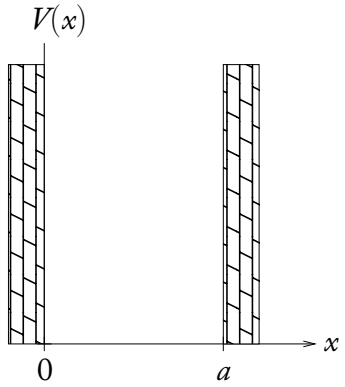


Figure 2.2.1: Infinite square well potential.

Because, the particle cannot have infinite energy, it cannot be found in the regions $x < 0$ and $x > a$. Therefore,

$$\psi(x) = 0, \quad \text{for } x < 0 \text{and } x > a. \tag{2.2.1}$$

For the region in the well, $0 \leq x \leq a$, we have $V(x) = 0$ and the Schrödinger equation reduces to

$$\psi''(x) + k^2\psi(x) = 0, \tag{2.2.2}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}} \tag{2.2.3}$$

and $E > 0$. We can write the general solution in terms of linear combinations of real solutions $\sin kx$ and $\cos kx$ as

$$\psi(x) = A \sin kx + B \cos kx. \tag{2.2.4}$$

Applying the boundary condition on the continuity of $\psi(x)$ at $x = 0$ sets $\psi(0) = B = 0$. We then have $\psi(x) = A \sin kx$, and applying the boundary condition at $x = a$ sets $\psi(a) = A \sin ka = 0$. Therefore, either $A = 0$ or $ka = n\pi$, $n = 1, 2, ...$ But $A = 0$ sets $\psi(x) = 0$ everywhere, which corresponds to no particle anywhere. Therefore,

$$k_n = \frac{n\pi}{a}, \quad n = 1, 2, ... \tag{2.2.5}$$

The wave vector is discrete, which means that both the momentum and the energy of the particle are discrete too, since $p = \hbar k$ and $E = \sqrt{\frac{2mE}{\hbar^2}}$. Therefore, $\psi_n(x) = A_n \sin k_n x = A_n \sin(\frac{n\pi}{a}x)$. Normalizing and choosing the wavefunctions to be real, $1 = \int_0^a |\psi_n(x)|^2 dx = \int_0^a A_n^2 \sin^2(\frac{n\pi}{a}x)dx = A_n^2 \frac{a}{2}$, or $A_n = \sqrt{\frac{2}{a}}$. The corresponding energies are $E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$. Summarizing, the nor-

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malized wavefunctions and the corresponding energies for the particle of mass m in the given infinite square well are

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin(\frac{n\pi}{a}x), \quad E_n = \frac{\hbar^2\pi^2n^2}{2ma^2}, \qquad n = 1, 2, 3, ... \tag{2.2.6}$$

We ignore $n = 0$, since $\psi(x)$ is identically zero in this case. We ignore $n = -1, -2, ...$, since $\psi_{-n}(x) = -\psi_n(x)$, the two differ only by an overall sign and are physically identical with the same probability density. The wavefunctions and the corresponding energies are shown in the schematic figure below.

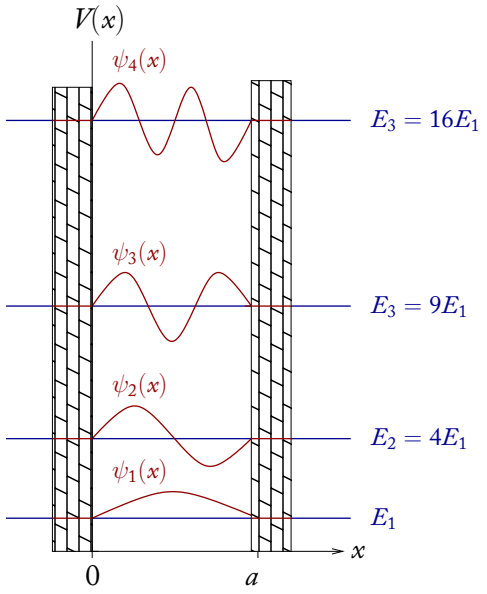


Figure 2.2.2: Wavefunctions and energies for a particle in the infinite square well potential.

Exercise 2.2.1:

1. Demonstrate that the particle cannot have $E = 0$ or $E < 0$ for the given potential well above the $x-$axis.
2. Show that $\psi(x) = 0$ outside the well by considering a finite potential of height V_0 and taking the limit $V_0 \to \infty$.

Solution:

1. If $E = 0$, then we have $\psi''(x) = 0$. This has the general solution $\psi(x) = Cx + D$. This with the boundary conditions $\psi(0) = \psi(a) = 0$ sets a straight line that passes through both $x = 0$ and $x = a$, or $\psi(x) = 0$, and there is no particle with

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$E = 0$. For $E < 0$, the Schrödinger equation becomes $\psi''(x) - \kappa^2\psi(x) = 0$, where $\kappa = \sqrt{-\frac{2mE}{\hbar^2}}$. The general solution is $\psi(x) = Ce^{-\kappa x} + De^{\kappa x}$. Applying the boundary conditions $\psi(0) = C + D = 0$ and $\psi(a) = Ce^{-\kappa a} + De^{\kappa a}$. Solving these, we get either $A = B = 0$ or $\kappa a = 0$. Therefore, we cannot have a particle with $E < 0$.

2. Considering a finite V_0, the general solution is $\psi(x) = Ce^{-\kappa x} + De^{\kappa x}$, where $\kappa = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$ now. Normalizability requires finiteness of the wavefunction at $x = \pm\infty$. Therefore, $\psi(x) = Ce^{-\kappa x}$ for $x > a$ and $\psi(x) = De^{\kappa x}$ for $x < 0$. Both vanish in the limit $V_0 \to \infty$ or $\kappa \to \infty$.

Exercise 2.2.2:

1. The wavefunctions and energies for a particle of mass m in a one-dimensional infinite square well with potential $V(x) = 0$ within $0 \le x \le a$ and $V(x) = \infty$ outside are given by $\psi_n(x) = \sqrt{\frac{2}{a}}\sin(\frac{n\pi}{a}x)$ for $0 \le x \le a$, $\psi_n(x) = 0$ outside, and $E_n = \frac{\hbar^2\pi^2n^2}{2ma^2}$, where $n = 1, 2, ...$

   (a) Determine $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, and Δx in state $\psi_n(x)$. Does Δx increase, decrease, or stay the same with increasing n?

   (b) Determine $\langle\hat{p}\rangle$, $\langle\hat{p}^2\rangle$, and Δp in state $\psi_n(x)$. Does Δp increase, decrease, or stay the same with increasing n?

   (c) Check whether the uncertainty principle holds for $\Delta x\Delta p$ in each state $\psi_n(x)$. Does $\Delta x\Delta p$ increase, decrease, or stay the same with increasing n? (5 points)

   (d) What is ΔE in state $\psi_n(x)$?

   (e) Determine $[\hat{x}, \hat{H}]$. Point out whether $\psi(x) = \hat{x}\psi_n(x)$ is a stationary state or not without explicitly determining $\psi(x, t)$.

   (f) The particle is in state $\psi(x, 0) = \psi_n(x)$ at time $t = 0$.

      i. What is the probability that the particle is found in the left half of the well within $0 \le x \le \frac{a}{2}$ at time $t = 0$?

      ii. What is the probability that the particle is found in the left quadrant of the well within $0 \le x \le \frac{a}{4}$ at time $t = 0$? Interpret and discuss the difference between even n and odd n.

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iii. What is the state $\psi(x,t)$ and the average energy of the particle at later time t? Do your answers for (1(f)i) and (1(f)ii) change at time t?

(g) The particle is in state $\psi(x,0) = c_1\psi_1(x) + c_2\psi_2(x)$ at time $t = 0$, where c_1 and c_2 are real constants.

i. Write down the normalized $\psi(x,0)$ and $\psi(x,t)$ at time t.

ii. List all possible values of energy that a single measurement could produce at time t and the corresponding probabilities. What is the expectation value of the energy (or average energy) of the particle at time t?

iii. Show that the probability density for finding the particle at point x within the well at time t oscillates in time. Determine the frequency and the amplitude of oscillation, and point out how the frequency is related to the energies E_1 and E_2.

iv. Determine $\langle\hat{x}\rangle$ at time t.

v. Determine $\langle\hat{p}\rangle$ at time t.

(h) The particle is in state $\psi(x,0) = \sqrt{\frac{30}{a^5}}\,x(a-x)$ for $0 \le x \le a$ and $\psi(x,0) = 0$ outside the well at time $t = 0$.

i. Make a schematic plot of $\psi(x,0)$. Determine the state $\psi(x,t)$ of the particle at later time t.

ii. What are the possible values that a single measurement of energy would produce at time t and the corresponding probabilities. Point out the energy with the highest probability and interpret and discuss the reason or what it means.

(i) Rewrite the wavefunctions when the location of the well where $V(x) = 0$ is moved to within $-\frac{a}{2} \le x \le \frac{a}{2}$.

Question 2.2.1: For the wavefunctions of the infinite square well, (A). $\psi(x)$ is continuous, (B). $\psi'(x)$ is continuous, (C). $\psi''(x)$ is continuous.

Question 2.2.2: For a single particle in a symmetric potential, the ground state wavefunction is (A). symmetric. (B). antisymmetric. (C). either symmetric or

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antisymmetric.

Question 2.2.3: How many nodes does $\psi_n(x)$ of the infinite square well have in the interior $0 < x < a$? (A). 0. (B). 2. (C). n. (D). $n-1$. (E). $n+1$. (F). It cannot be determined.

Question 2.2.4: Given stationary states $\psi_1(x,t)$ and $\psi_2(x,t)$, is $\psi_1(x,t)+\psi_2(x,t)$ stationary? (A). Yes. (B). No.

Question 2.2.5: Given $\psi_1(x,0) = x\psi_n(x)$ for the infinite square well, (A). $\psi(x,t) = x\psi_n(x)e^{-i\frac{E_n}{\hbar}t}$. (B). $\psi(x,t) = x\psi_n(x)e^{i\frac{E_n}{\hbar}t}$. (C). $\psi(x,t) = x\psi_n(x)$. (D). None of the above.

Question 2.2.6: Assuming that a quantum dot can be approximately described by an electron in a one-dimensional infinite square well potential, which solution of quantum dots has a well of largest width?



Figure 2.2.3: Quantum dots

(A). Red. (B). Orange. (C). Green. (D). Blue. (E). Violet.

2.3 Finite square well

Question 2.3.1: Considering a finite square well of the same width as the infinite square well with $V(x) = 0$ for $-\frac{a}{2} \leq x \leq \frac{a}{2}$ and $V(x) = V_0$ for $|x| > \frac{a}{2}$, which of the following are true about bound states in the finite square well? (A). The wavefunction is nonzero outside the well. (B). The ground state is symmetric with no nodes. (C). $\psi_n(x)$ has the same number of nodes as the nodes in the interior for $\psi_n(x)$ of the infinite square well. (D). E_n is smaller than E_n of the infinite square well. (E). There are a finite number of bound states.

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Now let us consider a particle of mass m in a potential well of finite depth,

$$V(x) = \begin{cases} -V_0, & -a \le x \le a \\ 0, & |x| > a. \end{cases}$$

The potential is shown in the figure below.

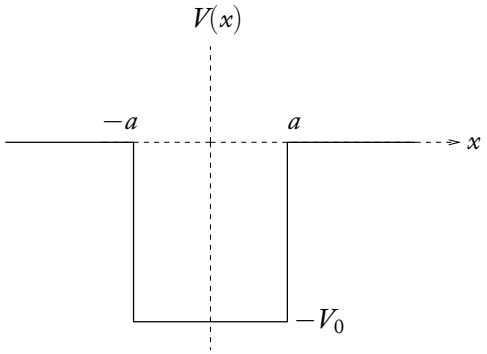


Figure 2.3.1: Finite square well.

Our interest here is to find the stationary bound states. Because the potential is piecewise and discontinuous, let us call the region on the left where $x < -a$ region I, that with in the well $a \le x \le a$ region II, and the region where $x > a$ region III. The Schrödinger equation in region I is

$$\psi_I''(x) - \kappa^2 \psi_I(x) = 0, \qquad \kappa \equiv \sqrt{-\frac{2mE}{\hbar^1}}, \tag{2.3.1}$$

where $E < 0$ for a bound state. The general solution is $\psi_I(x) = Ae^{-\kappa x} + Be^{\kappa x}$. But because the wavefunction has to vanish at $x = -\infty$, in order for it to be normalizable, we set $A = 0$, and we have

$$\psi_I(x) = Be^{\kappa x}. \tag{2.3.2}$$

The Schrödinger equation and the general solution in region III are exactly the same as in region I. The only difference is that the wavefunction in region III must vanish at $x = \infty$, which sets $B = 0$, and we have

$$\psi_{III}(x) = Ae^{-\kappa x}. \tag{2.3.3}$$

The Schrödinger equation in region II is

$$\psi_{II}''(x) + k^2 \psi_I(x) = 0, \qquad k \equiv \sqrt{\frac{2m(V_0 + E)}{\hbar^1}}, \tag{2.3.4}$$

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Because we can choose the solutions to be real and because we know that the bound states will be either even or odd, it is convenient to choose the general solution as a linear combination of sin and cos rather than complex exponentials, and we write

$$\psi_{II}(x) = C\sin kx + D\cos kx. \tag{2.3.5}$$

$C = 0$ and $D \neq 0$ correspond to even solutions, while $D = 0$ and $C \neq 0$. Let us investigate each case separately.

2.3.1 Even bound states

Even bound states with $\psi(-x) = \psi(x)$ pick out

$$\psi_{II}(x) = D\cos kx. \tag{2.3.6}$$

Because the potential has only a finite discontinuity at $x = \pm a$, both the wavefunction and its first derivative must be continuous at $x = \pm a$. Continuity of $\psi(x)$ at $x = a$, $\psi_{II}(a) = \psi_{III}(a)$, imposes

$$D\cos ka = Ae^{-\kappa a}. \tag{2.3.7}$$

Continuity of $\psi'(x)$ at $x = a$, $\psi'_{II}(a) = \psi'_{III}(a)$, imposes

$$-kD\sin ka = -\kappa Ae^{-\kappa a}. \tag{2.3.8}$$

Imposing continuity of $\psi(x)$ and $\psi'(x)$ at $x = -a$ does not add anything new; it reproduces the same constraints. Dividing the above two, we eliminate the constant coefficients, and we have

$$k\tan ka = \kappa. \tag{2.3.9}$$

Observe that given a and V_0, this is a constraint on the allowed energies of the bound states. However, this is a transcendental equation that does not have analytic solutions. We can solve it numerically or graphically as we discuss below. Using the definitions of k and κ and rewriting, $\tan ka = \frac{\kappa}{k}$, or $\tan\left(\sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}}\right) = \sqrt{\frac{-E}{V_0+E}} = \sqrt{\frac{-E-V_0+V_0}{V_0+E}} = \sqrt{-1+\frac{V_0}{V_0+E}}$; i.e.,

$$\tan\left(\sqrt{\frac{2m(V_0+E)a^2}{\hbar^1}}\right) = \sqrt{\frac{V_0}{V_0+E}-1}. \tag{2.3.10}$$

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Defining new variables

$$z \equiv \sqrt{\frac{2m(V_0 + E)a^2}{\hbar^2}}, \qquad z_0 \equiv \sqrt{\frac{2mV_0a^2}{\hbar^2}}, \tag{2.3.11}$$

the constraint becomes

$$\tan z = \sqrt{(\frac{z_0}{z})^2 - 1}. \tag{2.3.12}$$

The solutions can be obtained by plotting each side of (2.3.12) and finding the intersection points as shown in the figure below; i.e. the values of z for which both sides are equal.

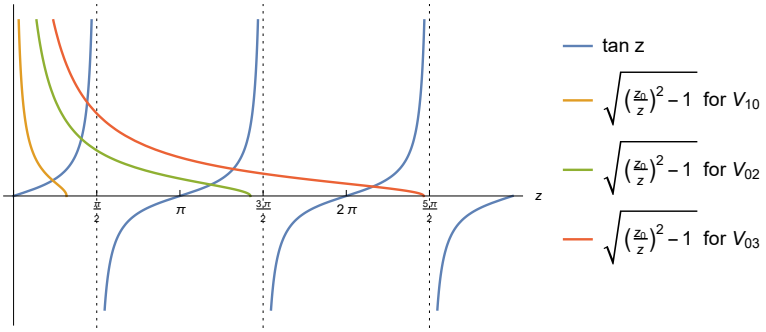


Figure 2.3.2: Graphic solution for the even bound states of the finite square well with depth $V_{01} < V_{02} < V_{03}$.

In the figure above, the potential V_{01} produces only one bound state, since there is only one intersection point, V_{02} produces two bound state, and V_{03} produces three bound state. We also see that any finite well in one dimension supports at least one bound state.

2.3.2 Odd bound states

For the odd states,

$$\psi_{II}(x) = C \sin kx. \tag{2.3.13}$$

Continuity of $\psi(x)$ at $x = -a$, $\psi_{II}(-a) = \psi_I(-a)$ imposes

$$C \sin ka = Be^{-\kappa a}. \tag{2.3.14}$$

Continuity of $\psi'(x)$ at $x = -a$, $\psi'_{II}(-a) = \psi'_I(-a)$, imposes

$$kC \cos ka = -\kappa Be^{-\kappa a}. \tag{2.3.15}$$

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Dividing the above two,

$$k \cot ka = -\kappa. \tag{2.3.16}$$

Rewriting this in terms of z and z_0 as in the even solutions, the constraint becomes

$$\cot z = -\sqrt{(\frac{z_0}{z})^2 - 1}. \tag{2.3.17}$$

This is also a transcendental equation which we solve using graphical method as shown in the figure below

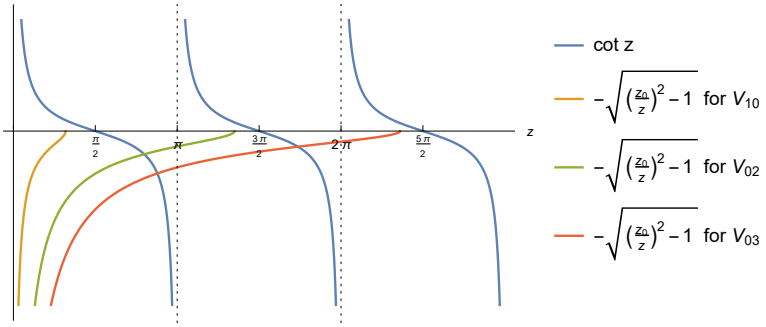


Figure 2.3.3: Graphic solution for the odd bound states of the finite square well with depth $V_{01} < V_{02} < V_{03}$.

In the figure above, the potential V_{01} does not support an odd bound state, since there is no intersection point, V_{02} produce only one bound state, and V_{03} produces two bound state. It means that a finite well potential supports at lease one even state, the ground state, and it may not support any of the excited states.

2.3.3 Wide and deep well

> **Question 2.3.2:** A wider well of the same depth supports (A). the same number of bound states. (B). larger number of bound states. (C). smaller number of bound states. (D). the same or less number of bound states. (E). the same or more number of bound states.

Now suppose the width and depth of the well are such that $z_0 = \sqrt{\frac{2mV_0a^2}{\hbar^2}} >> 1$. In this case, the curve $\sqrt{(\frac{z_0}{z})^2 - 1}$ for the odd case begins high above the $z-$axis and intersects $\tan z$ at its asymptotic values of $\frac{\pi}{2}$, $\frac{3\pi}{2}$, etc, as shown in the figure below.

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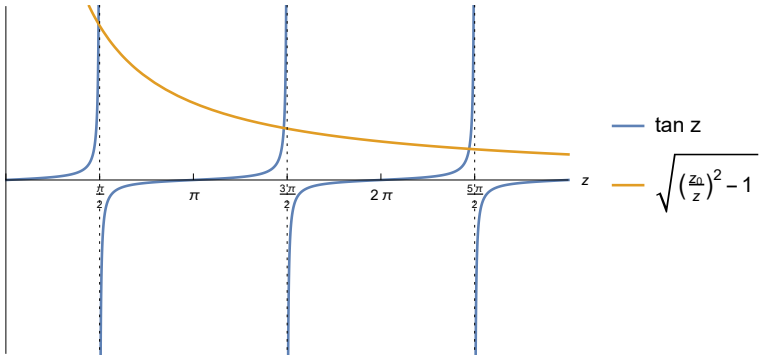


Figure 2.3.4: Graphic solution for the even bound states of a wide and deep finite square well.

Therefore, the intersection points for the odd states are at $z_n \approx \frac{n\pi}{2}$, n=1,3,5,... Similarly for the odd solutions, $z_n \approx \frac{n\pi}{2}$, $n = 2, 4, 6, \ldots$ Combining both the even and odd cases,

$$z_n \approx \frac{n\pi}{2}, \quad n = 1, 2, 3, \ldots \tag{2.3.18}$$

Using the definition of z, $\sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}} \approx \frac{n\pi}{2}$, and solving for E,

$$E_n = -V_0 + \frac{\hbar^2\pi^2 n^2}{8ma^2}, \quad n = 1, 2, 3, \ldots \tag{2.3.19}$$

Thus, the energy levels of the infinite square well of width $2a$ are reproduced for $E_n - (-V_0) = E_n + V_0$, since the ground state energy is shifted down and begins near $E = -V_0$ for the given potential.

2.3.4 Shallow and narrow well

We have already seen that any shallow or narrow well in one-dimension supports at least one bound state. But we need at least $z_0 = \sqrt{\frac{2mV_0a^2}{\hbar^2}} \geq \frac{\pi}{2}$ to support any odd state, as seen in Figure 2.3.3.

Question 2.3.3: Why is the ground state energy of the quantum oscillator (or any bound particle in a potential well) not zero, or why is the particle never at rest?

Question 2.3.4: Why are the energy levels of the finite square well smaller than the corresponding energy levels of the infinite square well?

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2.4 Delta function potential

Consider a particle of mass m in a delta function potential well,

$$V(x) = -\alpha\delta(x), \tag{2.4.1}$$

where α is a positive constant. A schematic figure of the potential is shown below.

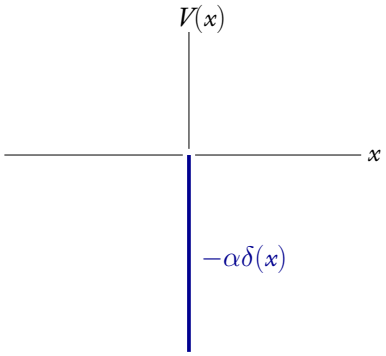


Figure 2.4.1: Delta function potential well.

To construct any bound state wavefunctions and the corresponding energies, we solve the Schrödinger equation on the left and the right sides of the potential and impose the continuity of $\psi(x)$ and the discontinuity of $\psi'(x)$ across it. Note that now that the potential has infinite discontinuity at the origin, $\psi'(x)$ is not continuous. Denoting the region $x < 0$ by I and the region $x > 0$ by II, the Schrödinger equation in region I is

$$\psi_I''(x) - \kappa^2\psi_I(x) = 0, \qquad \kappa \equiv \sqrt{-\frac{2mE}{\hbar^2}}, \tag{2.4.2}$$

where $E < 0$ for a bound state. The general solution is $\psi_I(x) = Ae^{-\kappa x} + Be^{\kappa x}$. But because the wavefunction has to vanish at $x = -\infty$, in order for it to be normalizable, we set $A = 0$, and we have

$$\psi_I(x) = Be^{\kappa x}. \tag{2.4.3}$$

The Schrödinger equation and the general solution in region II are exactly the same as in region I. The only difference is that the wavefunction in region II must vanish at $x = \infty$, which sets $B = 0$, and we have

$$\psi_{II}(x) = Ae^{-\kappa x}. \tag{2.4.4}$$

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Continuity of the wavefunction across $x = 0$ imposes

$$A = B. \tag{2.4.5}$$

The discontinuity of $\psi'(x)$ across $x = 0$ follows from

$$\begin{aligned} \psi'_{II}(0) - \psi'_{I}(0) &= \lim_{\epsilon\to 0}\int_{-\epsilon}^{\epsilon}\psi''(x)dx = \lim_{\epsilon\to 0}\int_{-\epsilon}^{\epsilon}\frac{2m}{\hbar^2}(V(x)-E)\psi(x)dx \\ &= -\frac{2m}{\hbar^2}\int_{-\epsilon}^{+\epsilon}\alpha\delta(x)\psi(x)dx = -\frac{2m}{\hbar^2}\alpha\psi(0) = -\frac{2m}{\hbar^2}\alpha A, \end{aligned} \tag{2.4.6}$$

On the other hand, taking the derivatives directly, $\psi'_{II}(0) - \psi'_{I}(0) = -\kappa A - \kappa B$. Therefore,

$$\kappa A + \kappa B = \frac{2m}{\hbar^2}\alpha A. \tag{2.4.7}$$

Solving (2.4.5) and (2.4.7), we have

$$\kappa = \frac{m\alpha}{\hbar^2}, \tag{2.4.8}$$

which using the definition of κ gives

$$E = -\frac{m\alpha^2}{2\hbar^2}. \tag{2.4.9}$$

Because there is only one solution, the delta function potential well supports one and only one bound state. Because $A = B$, the wavefunction decays symmetrically and exponentially on both sides. Writing $\psi_I(x) = Ae^{\kappa x}$ and $\psi_{II}(x) = Ae^{\kappa x}$, and normalizing, $1 = \int_{-\infty}^{0} A^2 e^{2\kappa x} + \int_{0}^{\infty} A^2 e^{-2\kappa x} = A^2\frac{1}{2\kappa} + A^2\frac{1}{2\kappa} = \frac{A^2}{\kappa}$, which gives $A = \sqrt{\kappa}$. Therefore, the normalized wavefunction is

$$\psi(x) = \begin{cases} \sqrt{\kappa}e^{-\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{-\frac{m\alpha}{\hbar^2}x}, & x > 0 \\ \sqrt{\kappa}e^{\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{\frac{m\alpha}{\hbar^2}x}, & x < 0. \end{cases}$$

The energy level and the wavefunction are shown in the schematic figure below.

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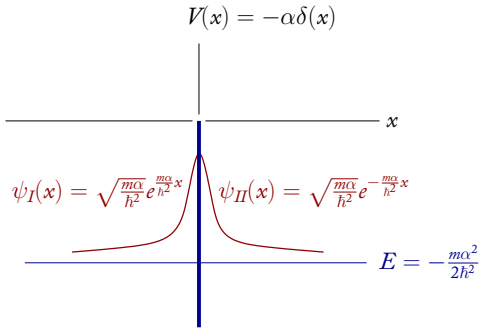


Figure 2.4.2: Energy and wavefunction of the delta function potential well.

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